

CONEPRO · DESIGN DECISION RECORD

ConePro TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CNP-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/conepro		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	01 decided by Amish on 2026-10-02 (recommendation approved, CNP-DEC-001)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); TRL 3 items now decided in CNP-DDR-002
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D8; item O1 was decided on 2026-10-02 (CNP-DEC-001)

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", and the design precis CNP-PRC-001 v0.2 listed the same key design choices with options. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. Items without a recommendation stay open.

The same instruction approved three portfolio-wide SwapCell decisions (a wake method for hosts without CAN, a charge-while-discharging mode and a vehicle latch vibration rating in SwapCell interface v0.3; shared packs priced once and excluded from dependent kit budgets) and directed that community designs pick co-design partners per area later. ConePro runs on three AA cells and does not use a SwapCell pack, so the SwapCell items do not apply to it.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in CNP-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Verticality (R6)	Option (a): a 3-axis accelerometer in the anvil sensor pad reads rod tilt between blows, with option (b), a bubble level on the handle, as a backup. Decided by Amish, 2026-09-25: go with recommendation.
D2	Rod extraction	Option (a): an optional lever rod puller with a rod clamp, carried separately and kept out of the instrument BOM total (R13) and the carried mass (R10). Decided by Amish, 2026-09-25: go with recommendation.
D3	Complete instrument or retrofit kit	Design the sensor set (BOM items 8 to 12) to fit standard 16 mm ASTM D6951 penetrometers from the start, and build the complete instrument for the prototype. The pitch in <code>project.yaml</code> is unchanged; the review made no rewording recommendation. Decided by Amish, 2026-09-25: go with recommendation.
D4	Depth sensing	Draw-wire sensor referenced to a plate on the ground, with a laser time-of-flight sensor as the fallback to compare later. Decided by Amish, 2026-09-25: go with recommendation.
D5	Hammer	Standard 8 kg hammer and 575 mm drop; the 4.6 kg option is noted for later. Decided by Amish, 2026-09-25: go with recommendation.
D6	Blow detection	Hall-effect sensor with the depth signal as a cross-check. With D1, the accelerometer's impact peak is added as a third signal. Decided by Amish, 2026-09-25: go with recommendation.
D7	Electronics location	Logger on the stationary plate, so only the sensor pad sees impacts. Decided by Amish, 2026-09-25: go with recommendation.
D8	Cells	Three AA cells rather than a lithium-ion cell. Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Item left open then, decided on 2026-10-02.

#	Item	Status
O1	First co-design partner and user group (small contractors, an NGO shelter or water team, or a low-volume road agency)	Decided by Amish, 2026-10-02 (recommendation approved): first partner to approach is a low-volume road agency (a county or district road department) that already runs manual DCP tests on unpaved roads; a university pavement or geotechnical lab is the second choice (CNP-DEC-001).

■ Consequences

- `project.yaml`: `budget_usd` stays \$400 (no change was recommended). Pitch and problem lines are unchanged.
- CNP-PRB-001, CNP-PRC-001 and CNP-REQ-001 are revised to v0.3. R6 is now met by design (D1). R9 now assumes the lever puller (D2). A new requirement, R15, carries D3: the sensor set must fit standard DCPs without machining. No target is relaxed.
- The BOM gains the accelerometer (line 10), the bubble level (line 6) and the optional extraction lever (line 15, excluded from the instrument total). The anvil loses its sensor pocket, because the pad now clamps on (D3).
- CNP-CAL-001 checks every requirement against the decided design. The new issues it raised (R10 mass with the bag, the R5 extension-rod clause, the draw-wire reel details and stiff-ground extraction) are now Decided by Amish, 2026-09-25: go with recommendation, and recorded in CNP-DDR-002 (D9 to D12).